



EXPLORATORY DATA ANALYSIS OF STUDENT ACADEMIC PERFORMANCE USING PYTHON

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SIMATS ENGINEERING



Engineer to Excel



ABSTRACT

This project performs Exploratory Data Analysis (EDA) on student academic performance data using Python. The objective is to uncover patterns, correlations and key factors that influence student performance. Using libraries such as Pandas, NumPy, Matplotlib and Seaborn, we clean, analyze and visualize data to derive meaningful insights that can help educators and institutions in decision making and student support strategies.



PROBLEM STATEMENT

- Educational institutions generate massive amounts of student data.
- It is challenging to manually analyze factors affecting academic performance.
- There is a need for data driven insights to improve learning outcomes.
- EDA helps in understanding data distributions, relationships and outliers before building models.



AIM

To explore and analyze student academic performance data using Python to identify key patterns, trends and factors that influence student outcomes.



OBJECTIVES

- Understand the dataset and its variables.
- Clean and preprocess the data for analysis.
- Perform univariate, bivariate and multivariate analysis.
- Visualize relationships and distributions.
- Derive actionable insights and recommendations.



DATASET OVERVIEW

- Source** : Kaggle – Student Performance Dataset (Student_Exam_Performance.csv)
- Total Records** : 1000
- Total Features** : 11
- Target Variable** : Performance Index (calculated)

Feature	Description
gender	Gender of the student (Male / Female)
race/ethnicity	Ethnicity group
parental level of education	Education level of parents
lunch	Lunch type (standard / free/reduced)
test preparation course	Completed test prep course (none / completed)
math score	Score in Mathematics (0-100)
reading score	Score in Reading (0-100)
writing score	Score in Writing (0-100)
hours studied per week	Number of hours studied weekly
absences	Number of absences
Performance Index	Average of three scores (Derived)



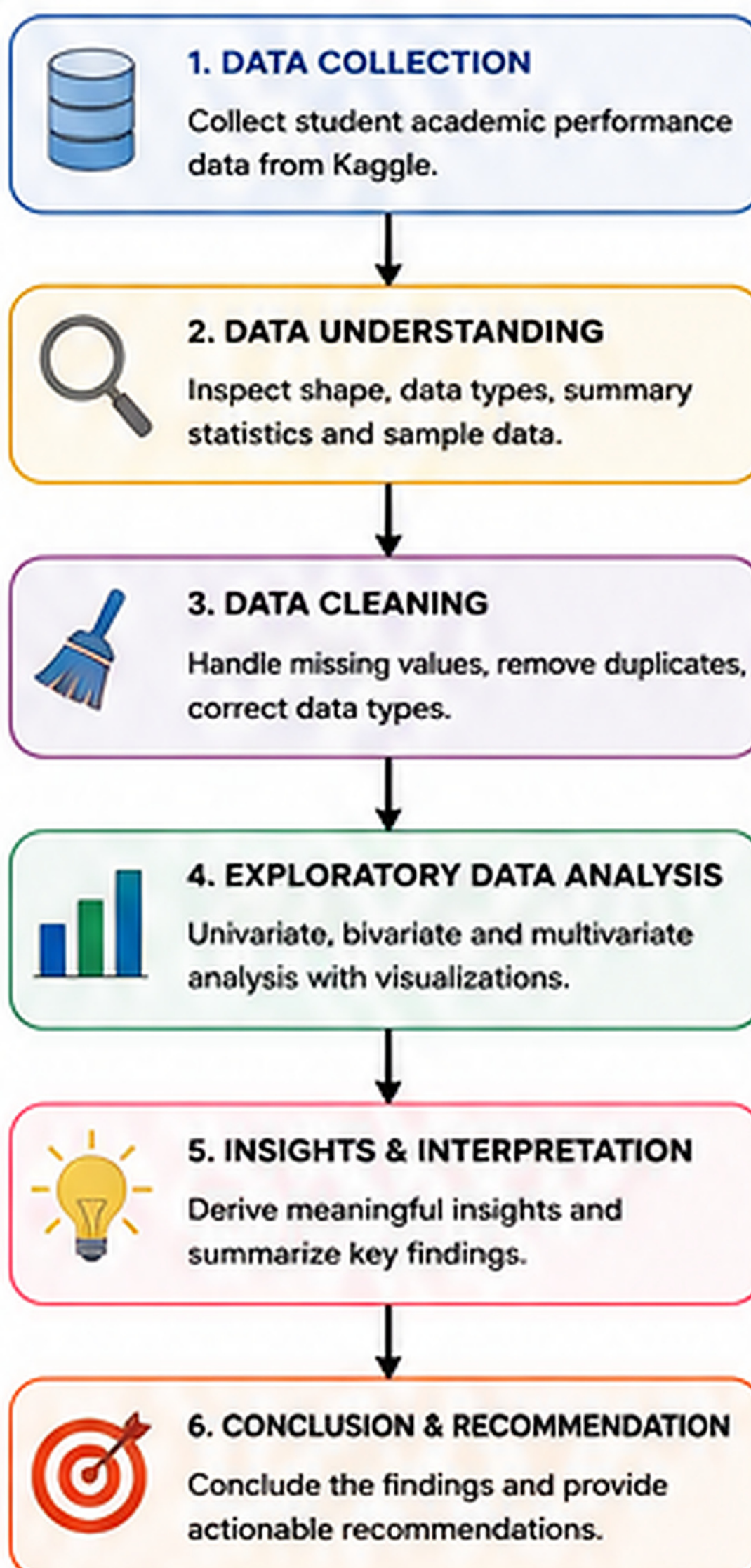
TOOLS & LIBRARIES



- pandas** : Data manipulation and analysis
- NumPy** : Numerical operations and arrays
- matplotlib** : Data visualization
- seaborn** : Statistical data visualization



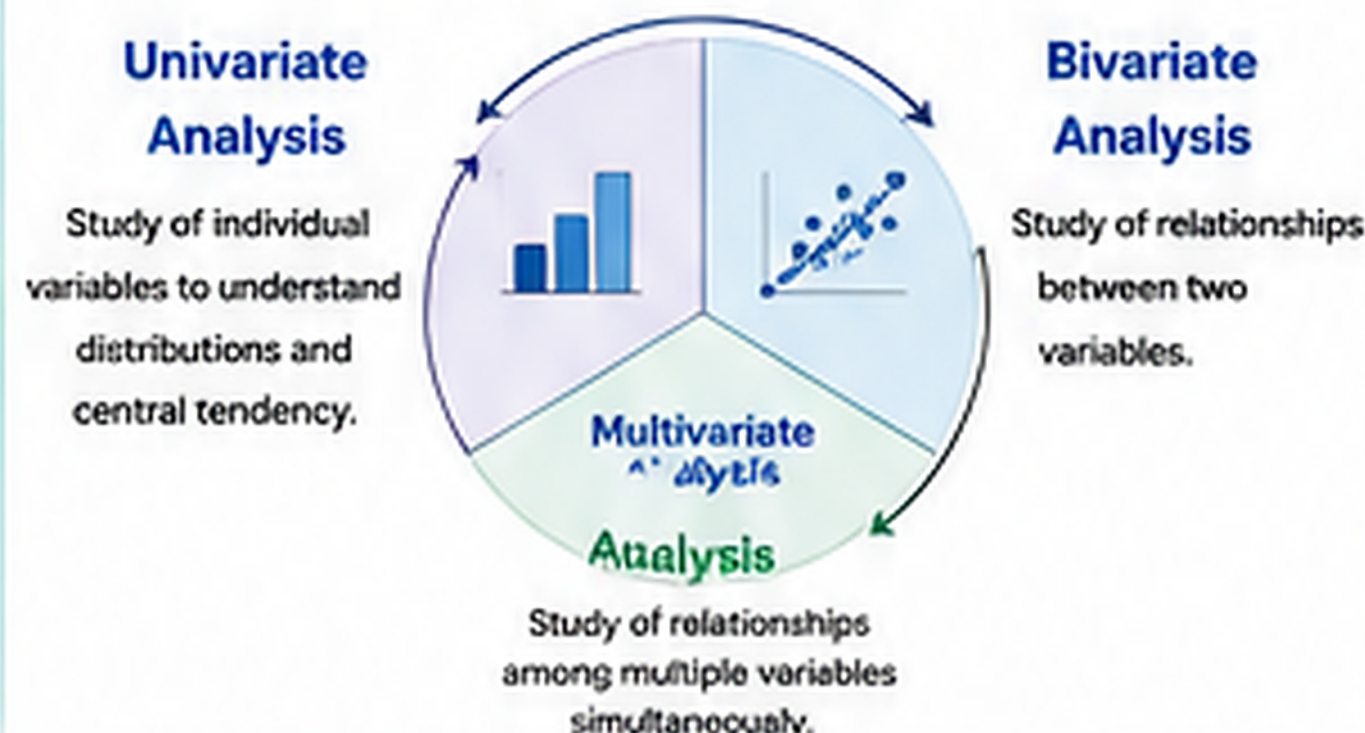
METHODOLOGY / EDA WORKFLOW



CORE EDA OPERATIONS

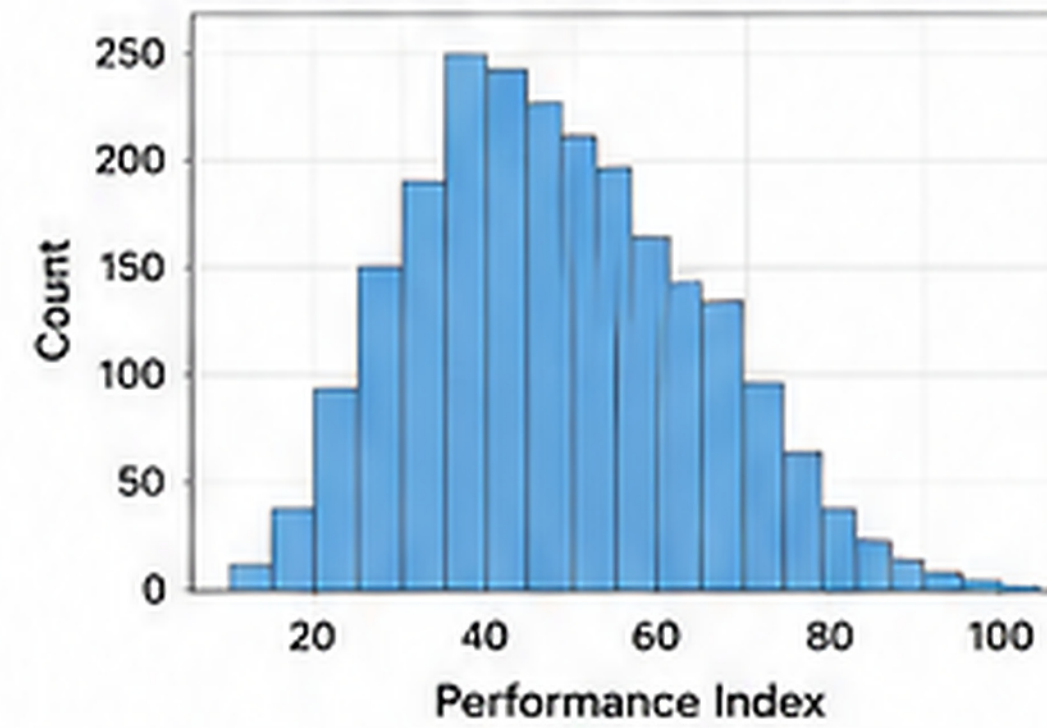
- Data Loading**
`df = pd.read_csv('Student_Exam_Performance.csv')`
- Data Inspection**
`df.head()`, `df.info()`, `df.describe()`
- Missing Values Analysis**
`df.isnull().sum()`
- Duplicate Records**
`df.duplicated().sum()`
- Data Cleaning**
`df.dropna()`, `df.drop_duplicates()`
- New Feature Creation**
`df['Performance Index'] = (df['math score'] + df['reading score'] + df['writing score']) / 3`
- Correlation Analysis**
`df.corr(numeric_only=True)`
- Data Visualization**
`matplotlib.pyplot`, `seaborn`

TYPES OF EDA PERFORMED



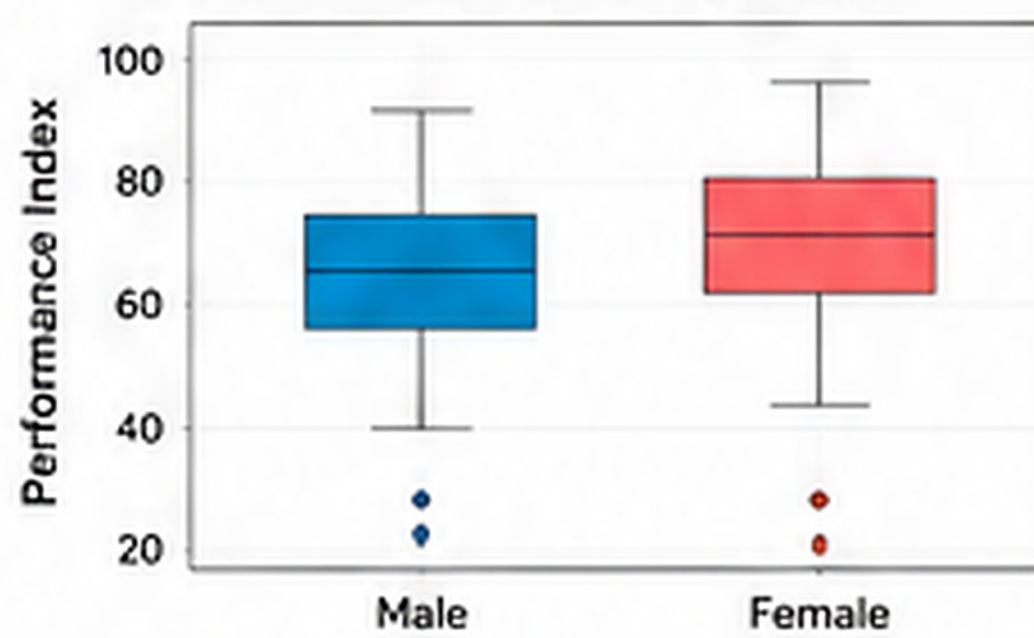
VISUALIZATION & KEY FINDINGS

1. Distribution of Performance Index



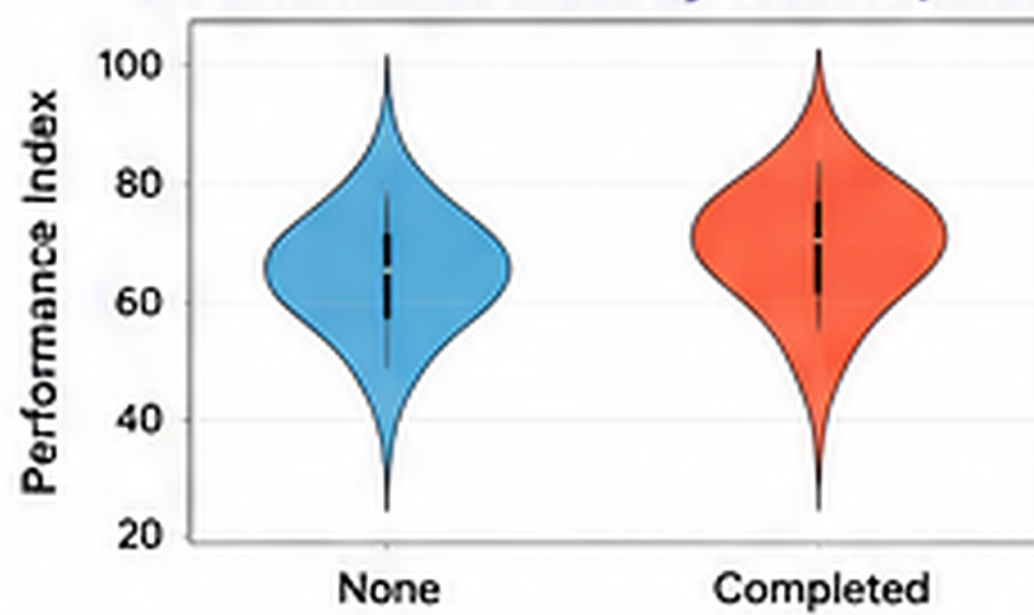
- Most students have performance index between 60 and 80. The distribution is slightly left-skewed.

2. Performance Index by Gender



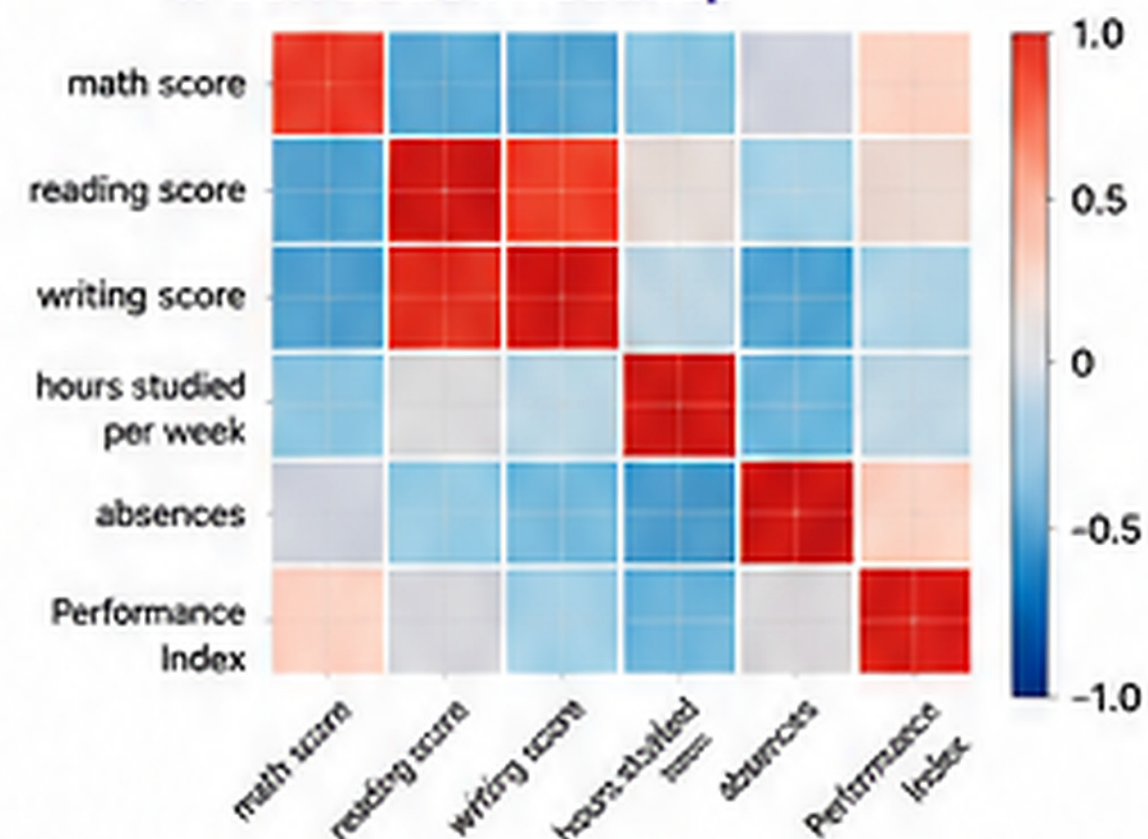
- Female students show slightly higher median performance than male students.

3. Performance Index by Test Preparation Course



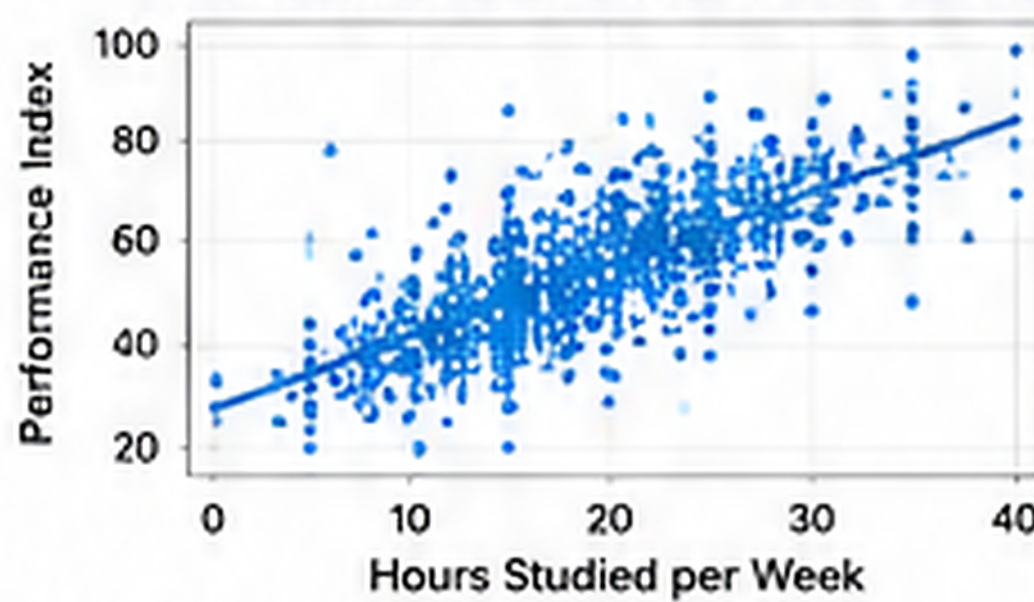
- Students who completed the test preparation course perform significantly better.

4. Correlation Heatmap



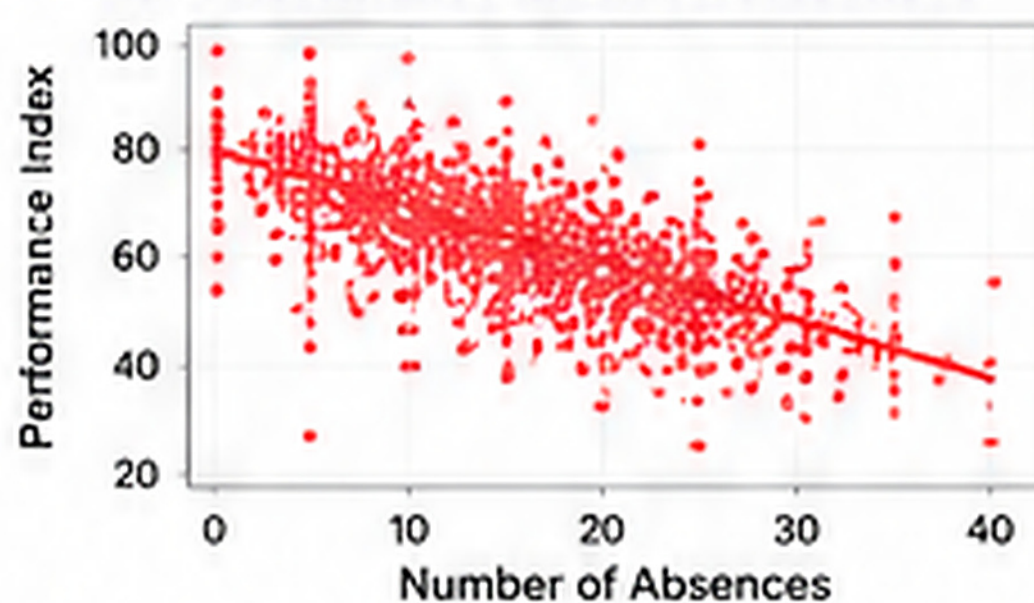
- Math, Reading and Writing scores are strongly positively correlated with Performance Index.
- Absences have negative correlation.

5. Performance Index vs. Hours Studied



- As study hours increase, performance index also increases. Positive linear relationship.

6. Performance Index vs. Absences



- More absences lead to lower performance index. Clear negative relationship.



INTERPRETATION PRINCIPLES

- Look at the distribution before making any assumptions.
- Compare groups to identify significant differences.
- Analyze correlations carefully; correlation does not imply causation.
- Combine multiple visualizations for deeper understanding.



KEY TAKEAWAYS

- Study hours positively impact performance.
- Test preparation course improves results.
- Absences negatively affect performance.
- Reading and writing scores are strong indicators of overall performance.
- Females show slightly better performance.



CONCLUSION

EDA reveals important trends and relationships in student academic performance. Insights from data help institutions to design better academic support programs and improve student success.



FUTURE WORK

- Build predictive models to predict performance.
- Include more features like parental income,

REFERENCES

- Kaggle – Student Performance Dataset
- pandas.pydata.org
- numpy.org
- matplotlib.org

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